

Lab: The Body Thinks Too

Objective

Explore how cognition is shaped by bodily states through exercises that reveal somatic markers, posture-thought connections, and the body's role in decision-making. This lab demonstrates that thinking is not confined to the brain.

Prerequisites

- A quiet space for 25 minutes
- Ability to stand, sit, and change posture
- A simple decision to make (e.g., what to eat, what to do this weekend, a small purchase)
- No prior knowledge required

Part 1: Posture and Thought (5 minutes)

1. **Collapsed posture:** Slump forward, let your shoulders round, drop your head. Stay here for one minute. Notice the quality of your thinking. What kinds of thoughts arise? What is the emotional tone?
2. **Upright posture:** Now sit or stand tall. Lift your chest, lengthen your spine, let your shoulders drop back. Stay here for one minute. Notice how the quality of thinking changes.
3. **Expansive posture:** Open your arms wide, take up space. One minute. What happens to your thoughts now?

Embodied cognition research shows that posture influences mood, confidence, and cognitive style. The body is not just executing thoughts decided elsewhere; the body's configuration shapes what can be thought.

Write your response here

(Describe the quality of thought in each posture. How did your thinking change when your body changed?)

Part 2: The Somatic Marker Check-In (7 minutes)

Think of the decision you brought to this lab. Hold it in mind lightly, without trying to solve it.

1. **Option A:** Bring the first option to mind. Now scan your body. What do you notice in your chest? Your belly? Your throat? Your shoulders? Rate the felt sense on a simple scale: constriction vs. openness, heaviness vs. lightness, warmth vs. cold.
2. **Option B:** Now bring the second option to mind. Repeat the body scan. What shifts?
3. If there is a third option, repeat again.

Antonio Damasio's somatic marker hypothesis proposes that the body generates feeling-states that guide decision-making before conscious reasoning arrives. These felt signals are the body's contribution to cognition.

Write your response here

(For each option, describe the bodily sensations. Which option felt more open or expansive? Which felt more contracted? Did the body have a preference before your mind did?)

Part 3: Thinking with Your Hands (5 minutes)

1. Think about a problem you are working on (it can be abstract). Sit still with hands in your lap and think about it for one minute.

2. Now stand up and begin to gesture as you think about the same problem. Use your hands, move around, draw shapes in the air. Think about it for one minute.
3. Now pick up a pen and paper. Sketch, diagram, or doodle as you think about the problem. One minute.

Research on gesture and cognition shows that hand movement is not just an expression of thought -- it is a constituent part of thinking. The hands offload cognitive processing, explore spatial relationships, and generate new ideas.

Write your response here

(How did your thinking change across the three conditions? Did gesturing or drawing produce any insights that sitting still did not?)

Part 4: Breath and Cognitive Clarity (5 minutes)

1. Breathe normally and try to recall a list: the days of the week backward. Notice the effort.
2. Take five slow, deep breaths (inhale for 4 counts, exhale for 6 counts). Now try again: months of the year backward. Notice if the effort has changed.
3. Breathe rapidly and shallowly for 15 seconds (not to the point of discomfort). Now try: count backward from 100 by 7s. Notice how cognitive clarity shifts.

The breath directly modulates arousal, attention, and cognitive performance. In Active Inference terms, respiratory patterns change the precision weighting of interoceptive signals, which in turn affects the clarity of cognitive processing.

Write your response here

(Describe how each breathing pattern affected your ability to think. What does this tell you about the body's role in cognition?)

Part 5: Reflection

Write your response here

(Before this lab, where did you think thinking happened? Has that changed? How would you now describe the relationship between body and mind?)

Reflection Table

Embodied Cognition Dimension	What I Noticed	Active Inference Connection
Posture and thought		Body configuration shapes generative model
Somatic markers		Interoceptive signals guiding inference
Gesture and cognition		Motor system as cognitive resource
Breath and clarity		Precision weighting via autonomic state